

Talia Berler

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PROFESSIONAL SUMMARY

Data scientist and AI researcher with experience delivering analytics solutions in healthcare, climate resilience, education, and data visualization. Adept at NLP analysis, creating ML pipelines, building LLM-powered solutions, and leveraging AI for evidence-based decision-making. Passionate about translating technical insights into actionable strategy for cross-functional stakeholders.

TECHNICAL SKILLS

Languages & Web: Python, Java, R, JavaScript, C++, SQL, HTML, CSS

AI / LLM Technologies: Large Language Models APIs, Prompt Engineering, RAG Pipelines, Fine-Tuning

ML Frameworks: PyTorch, TensorFlow, Hugging Face Transformers, Scikit-learn

Data Wrangling & Geospatial: Pandas, NumPy, BeautifulSoup, GIS, Excel, Matplotlib, Tableau, Adobe Creative Suite

DevOps & Cloud: Git/GitHub, Azure, Docker, VS Code, Colab, JupyterLab

EXPERIENCE

CLIMATE RESILIENCE INSTITUTE – Coral Gables, FL

Sep 2025 – Present

Graduate Resilience Intern

- Building scalable Resilience Finance Repository MVP with AI enrichment features for 40+ stakeholders to support knowledge-sharing and knowledge gap identification goals of Resilience Finance Lab.
- Coordinated main stage conference operations for Resilience 365, managing materials and logistics to ensure seamless execution for 350+ attendees and 90+ industry-leading speakers.

UM LAW & AI LAB (MILA) – Coral Gables, FL

Jul 2024 – May 2026

ClassInsight Project Lead

- Lead student developer producing LLM-based agentic classroom-feedback application, ClassInsight
- Working with UI developer to formalize and scale beta app for publication of open-source code on GitHub

Research Assistant, Dev Ops Team

- Assisted PI in building a hub for interdisciplinary research, innovation, and education in the field of AI and law, provided students with AI-related literacy and skills within the legal domain and facilitate AI use in research and teaching for interested faculty
- Leveraged popular LLMs, including OpenAI and Anthropic models, to develop instructional tools at the University of Miami School of Law

MOFFITT CANCER CENTER, LAD VALDES – Tampa, FL

May 2025 – Nov 2025

Research Trainee

- Led a month-long experiment with an external consulting team to evaluate 6+ state-of-the-art retrieval algorithms for a medical RAG system of 20+ databases containing over 1 million text records
- Generated new benchmark datasets for training and evaluating a logistic regression model to classify specific types of cancer within the 283B tokens of cancer text data in The Blue Scrubs v2.0 with 692B tokens of medical text
- Collected and processed 500K+ clinical trial records and clinical guideline documents to build robust RAG databases for The Blue Scrubs Virtual Clinicians via scraping, API access, and public data downloads

VALIDIT.AI – Tel Aviv, IL

Jun 2023 – Jul 2023

Development Intern

- Executed rigorous quality assurance testing on trust verification application and admin site
- Revamped website to effectively market the application to investors and potential clients

SELECTED PROJECTS

Assessing Agricultural Land Suitability under Future Climate Scenarios, California's Central Valley

GIS II Final Project

- Built an ArcGIS-based multi-criteria evaluation model combining soil, climate, hydrology, elevation, and land-cover data to map agricultural suitability
- Compared baseline and future climate scenarios (SSP2-4.5, SSP5-8.5) for 2050, producing suitability and change maps highlighting gains, losses, and stable zones

ClassInsight – LLM-Driven Classroom-Feedback Application

MiLA Lab

- Developed agentic classroom-feedback system using GPT and Anthropic APIs; integrated retrieval workflows and prompt-engineering strategies
- Co-led UI collaboration and prepared open-source release; shortlisted finalist at AIREA 2025

Cancer-Type Classifier for The Blue Scrubs v2.0 Dataset

Moffitt Cancer Center

- Built a TF-IDF + Logistic Regression model to classify cancer types within a 692B-token clinical text corpus, creating a curated 30K-sample breast-cancer dataset from 11.5M records using keyword bootstrapping and sampling filters
- Achieved strong performance on hold-out and out-of-sample sets (F1 0.86, AUC 0.94) with <3-minute CPU training pipeline

Medical Retrieval-Augmented Generation System

Moffitt Cancer Center

- Evaluated 6+ retrieval algorithms across 1M+ medical text records; designed benchmarking datasets for classifier training
- Processed 500K+ clinical trial and guideline documents using scraping, APIs, and ML preprocessing pipelines

EDUCATION

Master of Science (M.Sc.), Data Science

SMART Cities & GIS Concentration; GPA: 4.00/4.00

University of Miami, Coral Gables, FL | May 2026

Bachelor of Science (B.Sc.), Computer Science

Magna Cum Laude; GPA: 3.96/4.00

University of Miami, Coral Gables, FL | Dec 2024

High School Diploma

Magna Cum Laude; GPA: 4.00/4.00

Ransom Everglades School, Miami, FL | May 2020

EXTRACURRICULARS

CLIMATE RESILIENCE INSTITUTE – Miami, FL

Dec 2024 – Present

Student Advocacy Council Member

- Contributing to bimonthly meetings of the Student Advocacy Council in order to plan events, programming, conferences, and advocacy strategy to encourage resiliency work on campus and in broader Miami
- Designing interdisciplinary projects to address climate resilience preparation in Miami with team members from across University of Miami's academic departments

BONSAI APPLIED COMPUTATIONS GROUP – Miami, FL

Sep 2024 – Mar 2025

Graduate Analyst, Product Team

- Assisted team in developing genetics-based product optimization algorithm with feature ranking and multi-generational product tracking
- Researched data visualization types and software to visualize evolution of products through life cycle

Undergraduate Analyst, Marketing Team

- Designed and implemented algorithms to perform sentiment analysis and produce time series visualizations of large political tweet datasets
- Collaborated with product managers and cross-functional teams to develop and optimize computational programs for research-focused projects, delivering sprint-based solutions for third-party clients such as companies and professors

CERTIFICATIONS

GIS for Climate Action — ESRI, issued Apr 2026

IBM Data Science Practitioner — IBM, issued Jun 2025

(Credential ID: IBMDesign20250604-28-opwtxt)

IBM Enterprise Design Thinking Practitioner — IBM, issued Apr 2025

(Credential ID: IBMDesign20250604-26-375fdi)

HONORS & AWARDS

Shortlisted Finalist, AIREA Competition, Hong Kong, Jul 2025

Phi Beta Kappa, May 2025

President's & Provost's Honor Rolls, 2022-2024

Dean's List, 2022 - 2024

Faculty Cup, Segall '82 Leadership Award, Ransom Everglades School, May 2020

PUBLICATIONS

Spiess, P. E., Zitu, M. M., Walker, A., Anaya, D. A., Wenham, R. M., Vogelbaum, M., Fernandez, C. G., Felipe, L., ElNaqa, I., Delgado, E., **Berler, T.**, & Valdes, G. (2026). *BPI26-015: Complementing NCCN guidelines with a clinician-guided ai reasoning platform: Evaluation of the BlueScrubs*. Journal of the National Comprehensive Cancer Network, 24(3.5). <https://doi.org/10.6004/jnccn.2025.7367>

Spiess, P. E., Zitu, M. M., Walker, A., Anaya, D. A., Wenham, R. M., Vogelbaum, M., Grass D., Jaffer A.-M., Sarnaik A., McMullen C., Sam C., Kiluk J. V., Liu T., Biachi T., Powsang J., Chern J.-Y., Li R., Felder S., Reynolds S., Shafique M., Sheehan A., Layman A., Warfield C. A., Legoas D., Parrinello J., Schmitz J., Eaton K., Honor M., Felipe L., ElNaqa I., Delgado E., **Berler T.**, Phillips R. V., Francisque F., Fernandez C. G., Valdes G. (2026). *Clinical Reasoning AI for Oncology Treatment Planning: A Multi-Specialty Case-Based Evaluation*. arXiv [Cs.CY]. <https://doi.org/10.48550/arXiv.2604.20869>

Berler, T., & Cohen-Sasson, O. (2026). *ClassInsight: Developing an Agentic AIED Tool for Responsive Teaching and Learning*. Research Square. <https://doi.org/10.21203/rs.3.rs-8078768/v1>

Shotande, M., Felipe, L., Garcia, C., **Berler, T. K.**, El Naqa, I., Tripathi, A., Rudapratna V., Rasool G., & Valdes, G. (2025). *The Blue Scrubs v2: A Comprehensive Curated Medical Dataset Derived from the Internet using FineWeb*. Retrieved from <https://huggingface.co/datasets/TheBlueScrubs/TheBlueScrubs-v2>

Zitu, M., Spiess, P., Walker, A., Anaya, D., Wenham, R., Vogelbaum, M., Garcia Fernandez, C., Felipe, L., El Naqa, I., Delgado, E., **Berler, T.**, & Valdes, G. (2025). *Clinical Artificial Reasoning in Oncology: Dawn of a Transformative Era*. SSRN. <https://doi.org/10.2139/ssrn.5649971>

CONFERENCE PRESENTATIONS

UNIVERSITY OF MIAMI GRADUATE RESEARCH SYMPOSIUM – Miami, FL

Mar 2025

Oral Presenter, "Class Insight: An LLM-Driven Agentic Classroom-Feedback Application" (15-min talk + Q&A)

ADDITIONAL INFORMATION

Languages: English, Proficient Spanish

Interests: Climate Advocacy, Environmental Conservation, Data Science for Social Impact, Journalism, Creative Writing